RECURSION

Recursion

1. A function, which calls itself either directly or indirectly through another function.
2. A recursive function is called to solve a problem.
3. The function actually knows how to solve only the simplest case(s), or so-called **base case(s)**
4. If the function is called with the **base case** it ***simply returns the result or in some cases do nothing***.

Recursion

5. If the function is called with a more complex problem (A), the function divides the problem into **two** conceptual pieces (B and C).

   These two pieces are a piece that the function **knows how to do** (B – a base case) and a piece that the function **does not know how to do** (C).

   The latter piece (C) must resemble the original problem, but be a slightly simpler or slightly smaller version of the original problem (A).

Recursion Guidelines

- The definition of a recursive method typically includes an `if-else` statement.
  - One branch represents a base case which can be solved directly (without recursion).
  - Another branch includes a recursive call to the method, but with a “simpler” or “smaller” set of arguments.
- Ultimately, a base case must be reached.

Rules of Recursion

- Bad use of recursion, causes a huge amount of redundant work being performed, violating a major rule of recursion.

Rules of Recursion:

1. **Base Case:** You must always have some base cases, which can be solved without recursion.
2. **Making Progress:** Recursive call must always be to a case that make progress toward a base case.

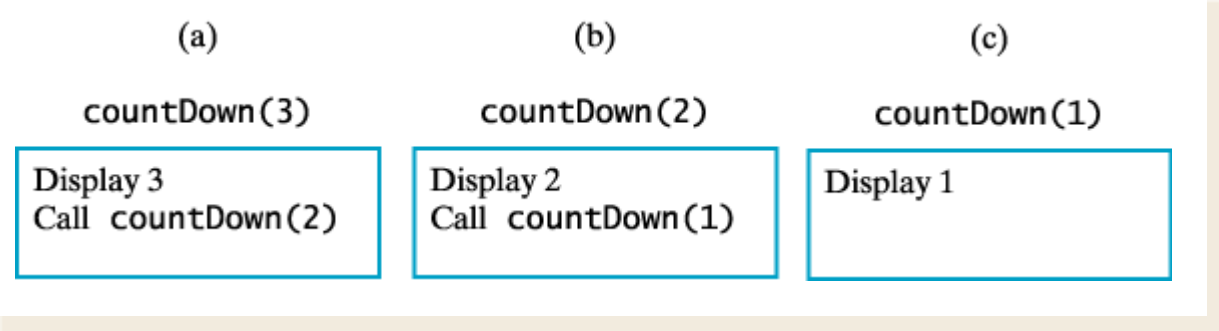
Infinite Recursion

- If the recursive invocation inside the method does not use a “simpler” or “smaller” parameter, a base case may never be reached.
- Such a method continues to call itself forever (or at least until the resources of the computer are exhausted as a consequence of *stack overflow*).
- This is called *infinite recursion*.

Tracing a Recursive Method

- Given:

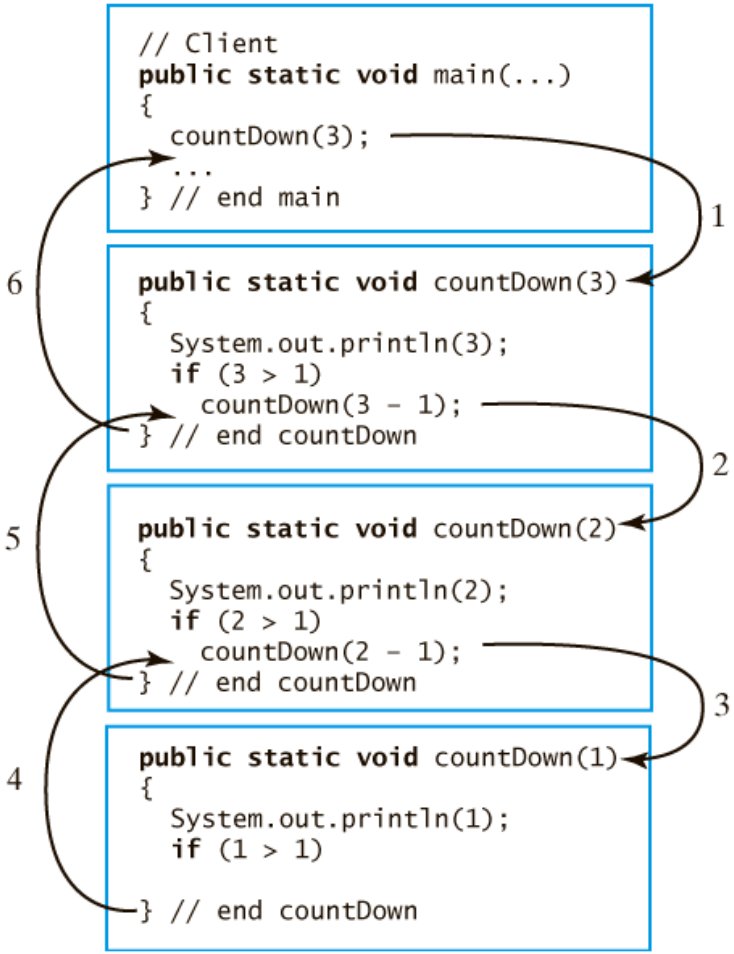
```
public static void countDown(int integer)
{   System.out.println(integer);
    if (integer > 1)
        countDown(integer - 1);
} // end countDown
```



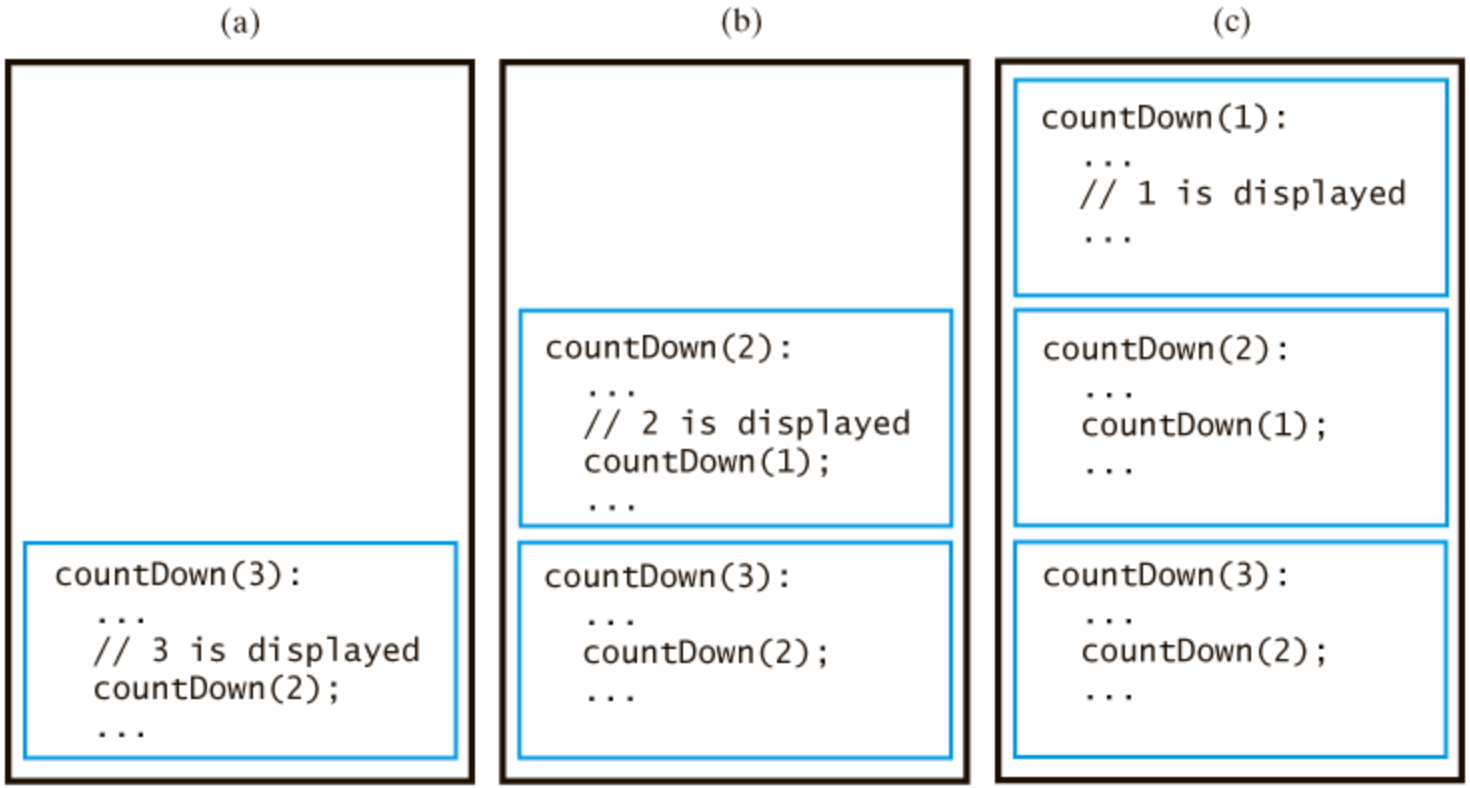
The effect of method call **`countDown(3)`**

Tracing a Recursive Method

Tracing the recursive call **`countDown(3)`**

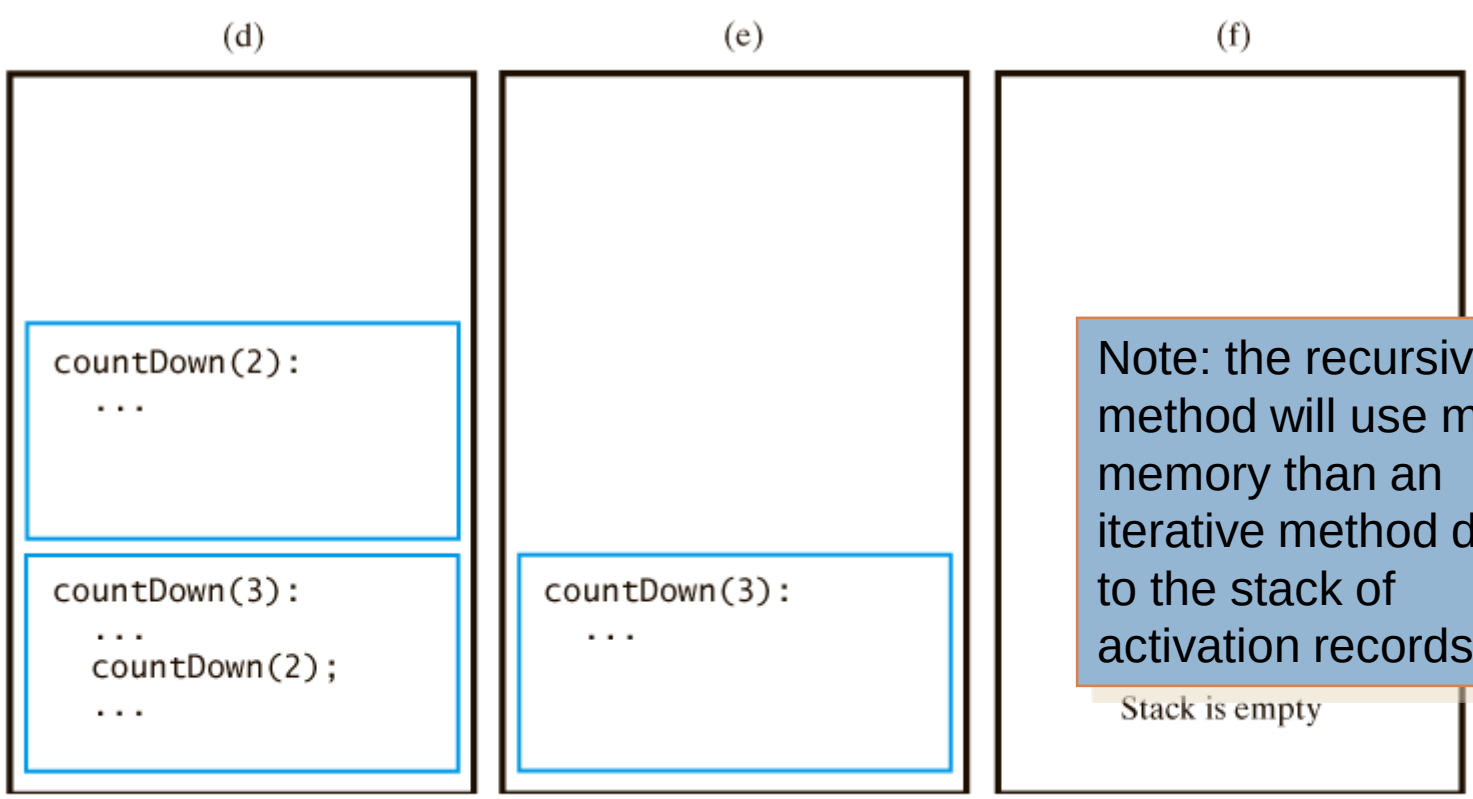


Tracing a Recursive Method



The stack of activation records during the execution of a call to **`countDown(3)`**… continued →

Tracing a Recursive Method



The stack of activation records during the execution of a call to **`countDown(3)`**

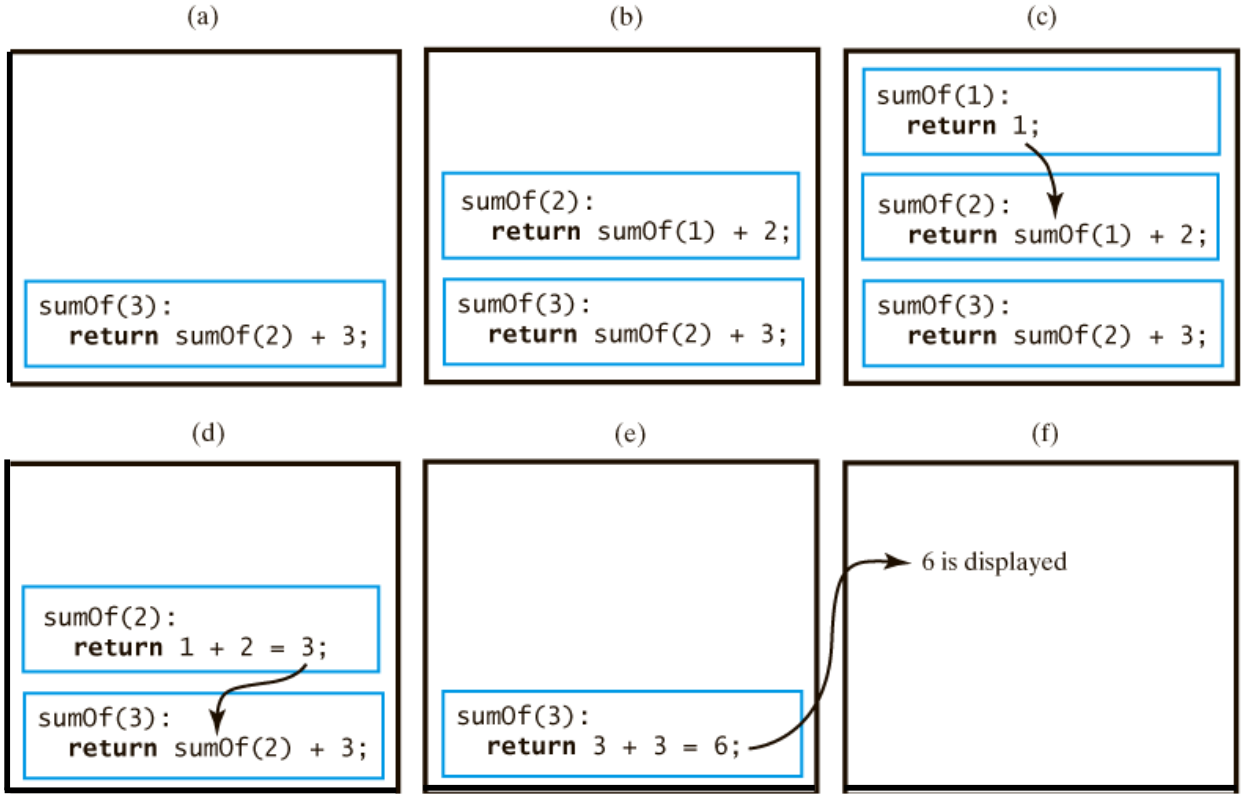
Recursive Methods That Return a Value

- Task: Compute the sum

  1 + 2 + 3 + … + n for an integer n > 0

```
public static int sumOf(int n)
{   int sum;
    if (n = = 1)
        sum = 1;                        // base case
    else
        sum = sumOf(n - 1) + n;  // recursive call
    return sum;
} // end sumOf
```

Recursive Methods That Return a Value



The stack of activation records
during the execution of a call to `sumOf(3)`

Recursion vs. Iteration

- Any recursive method can be rewritten without using recursion.
- Typically, a loop is used in place of the recursion.
- The resulting method is referred to as the *iterative version.*

Recursion vs. Iteration, cont.

- A recursive version of a method typically executes less efficiently than the corresponding iterative version.
- This is because the computer must keep track of the recursive calls and the suspended computations.
- However, it can be much easier to write a recursive method than it is to write a corresponding iterative method.

Recursion

- Recursion can describe everyday examples
  - Show everything in a folder and all it subfolders
    - show everything in top folder
    - show everything in each subfolder in the same manner

Recursive Methods That Return a Value

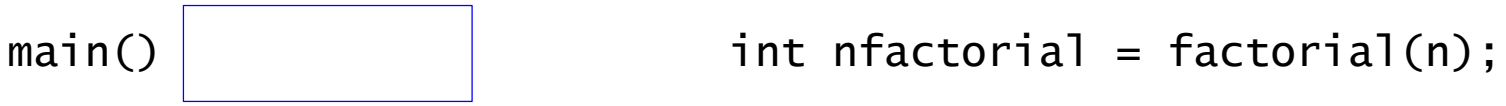
- A recursive method can be a `void` method or it can return a value.
- At least one branch inside the recursive method can compute and return a value by making a chain of recursive calls.

Method factorial()

```
public static int factorial(n) {
   if (n == 0) {
        return 1;
   }
   else {
        return n * factorial(n-1);
   }
}
```

Recursive invocation

- A new activation record is created for *every* method invocation
  - Including recursive invocations



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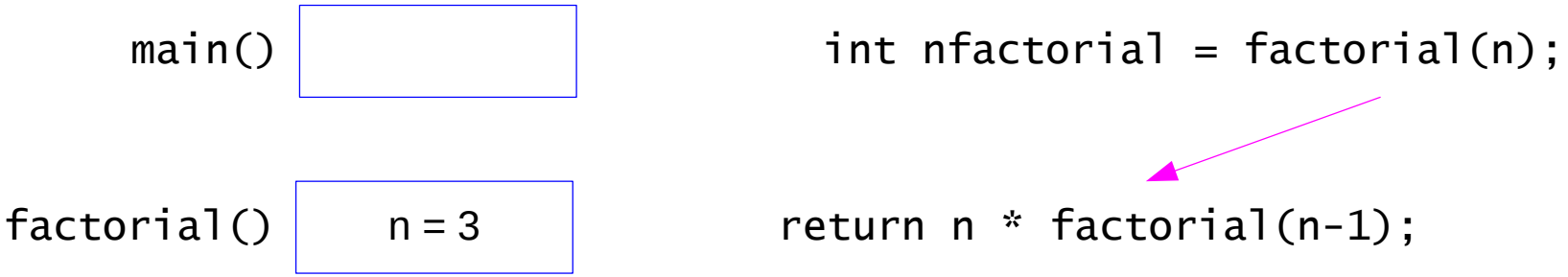

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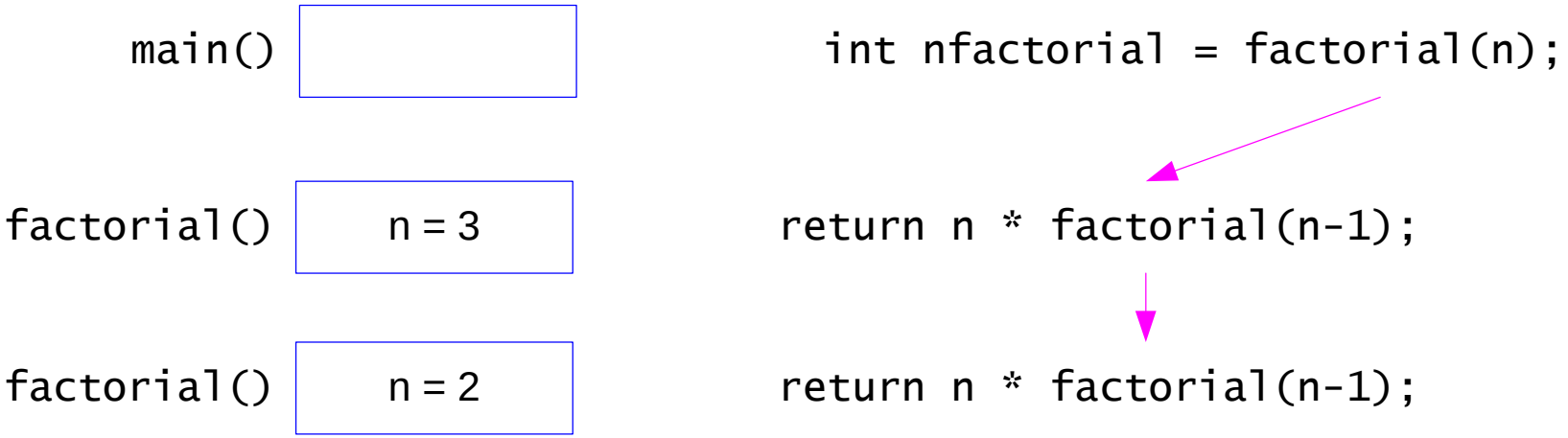

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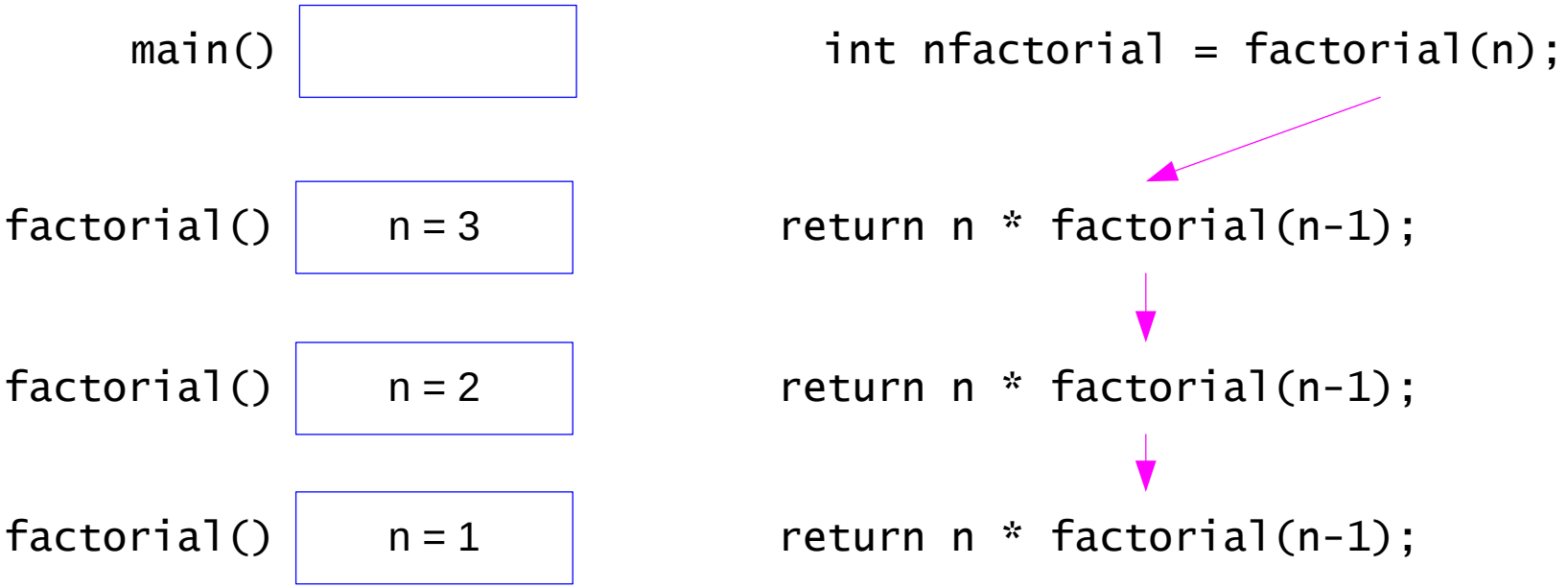

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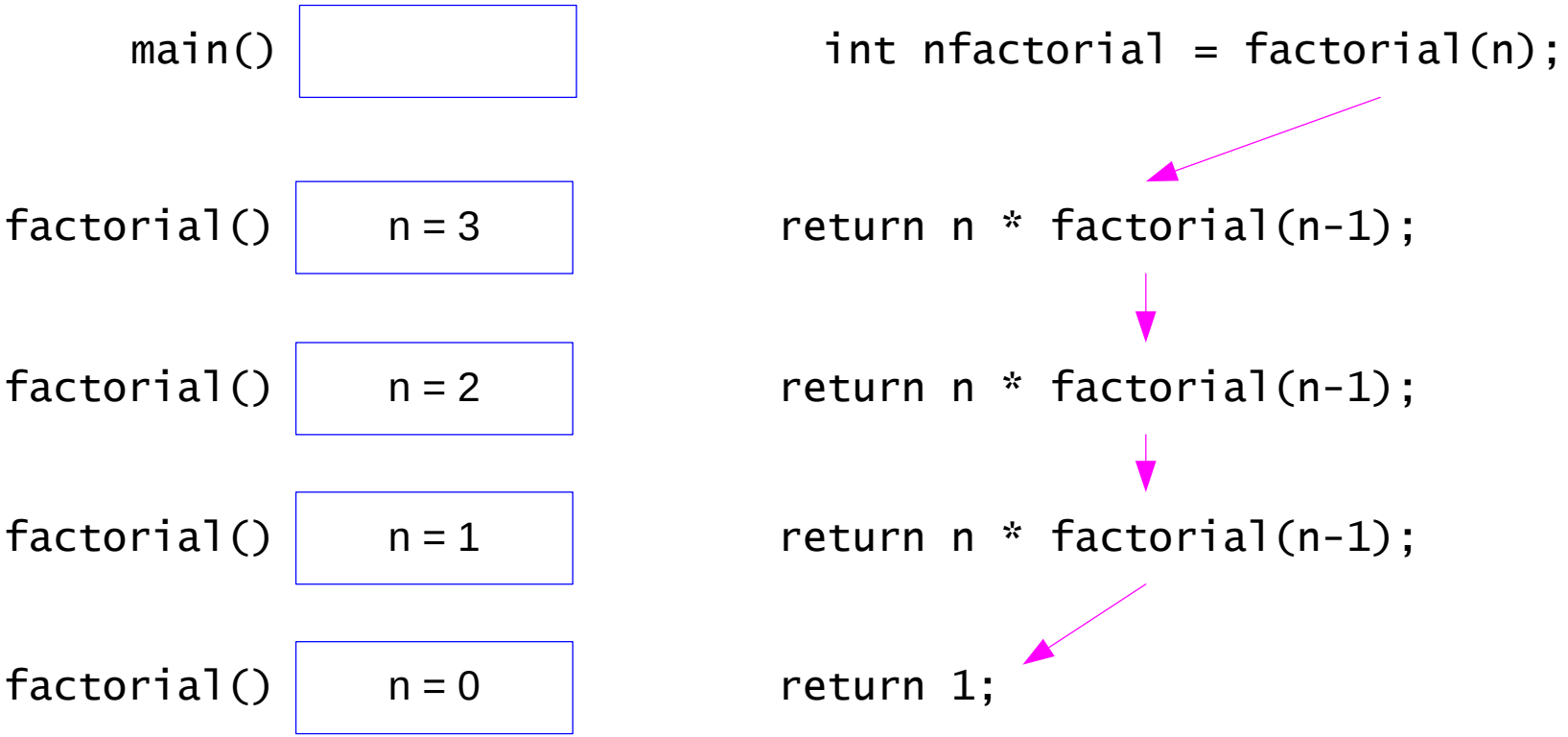

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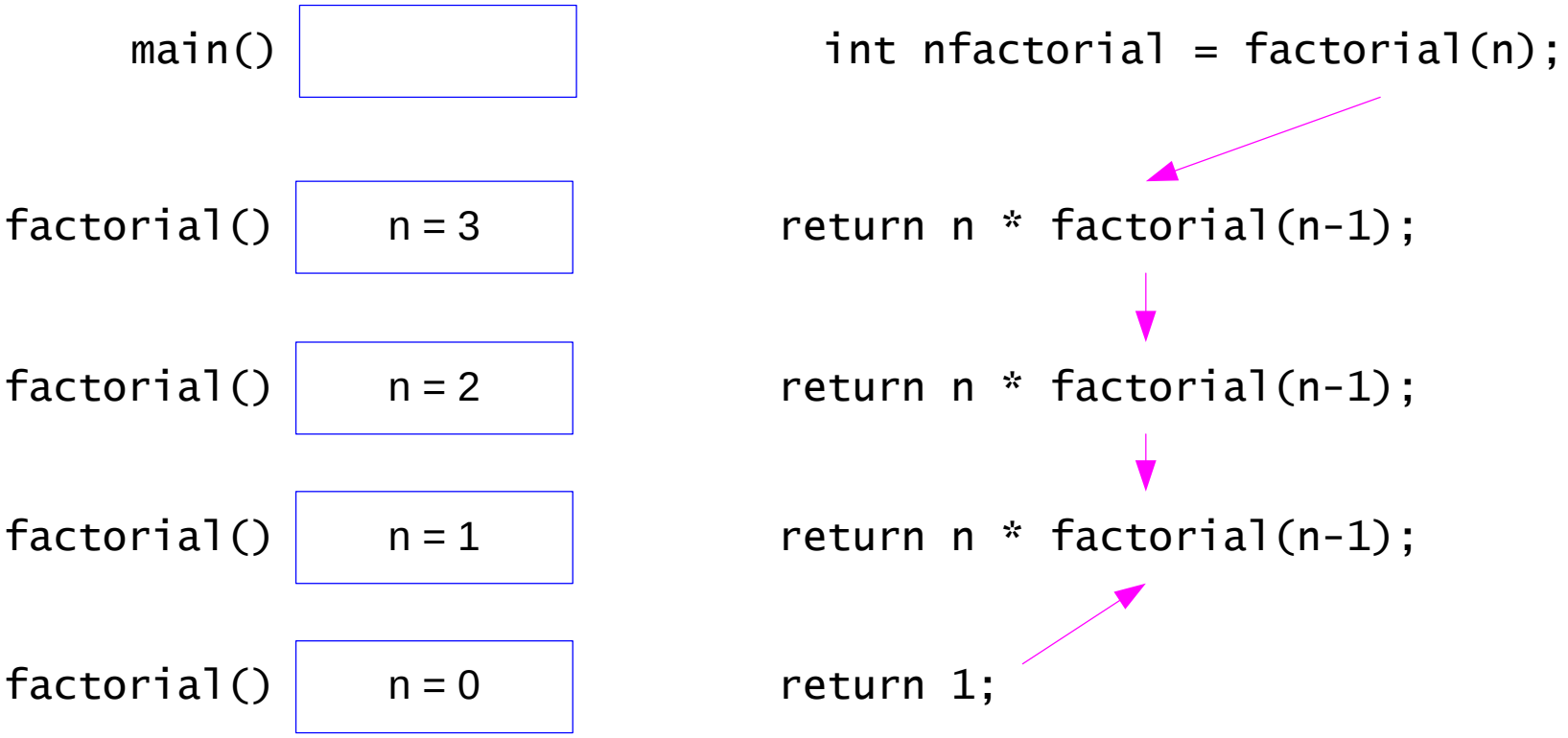

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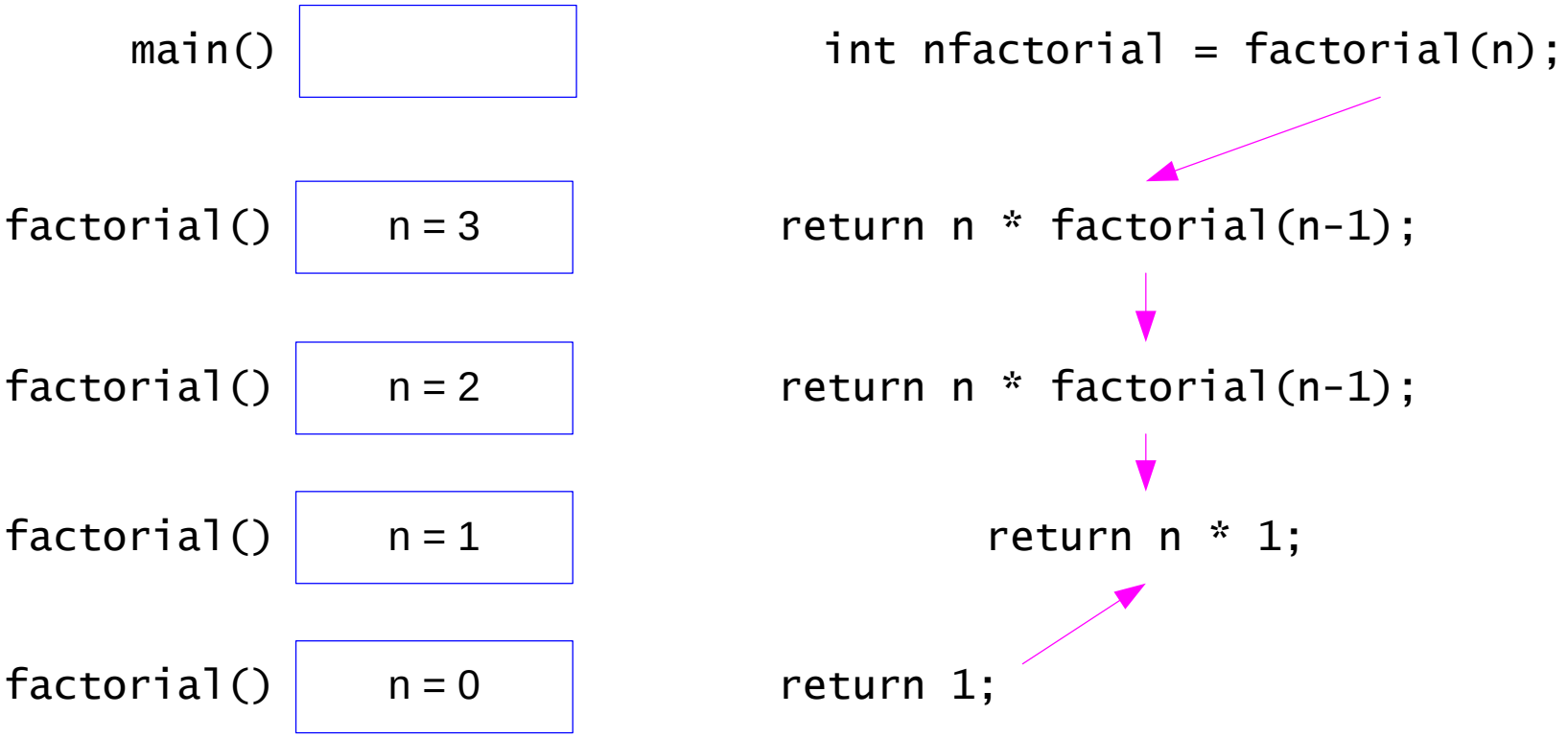

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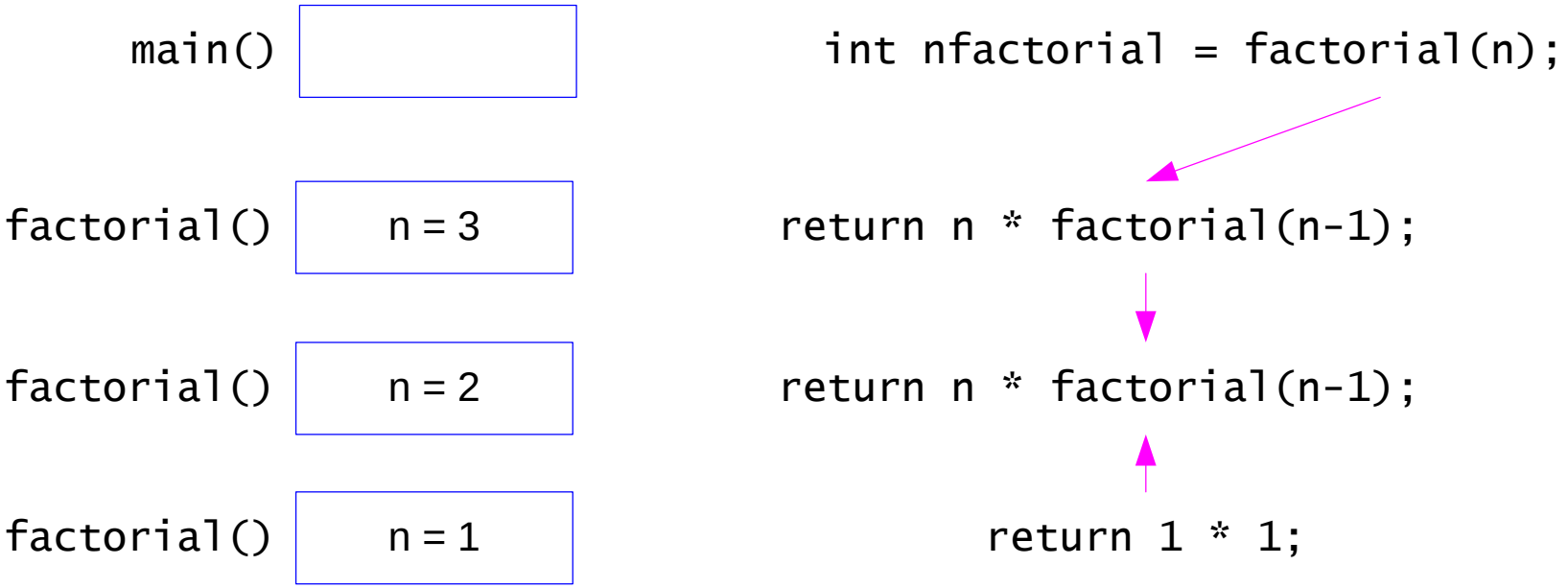

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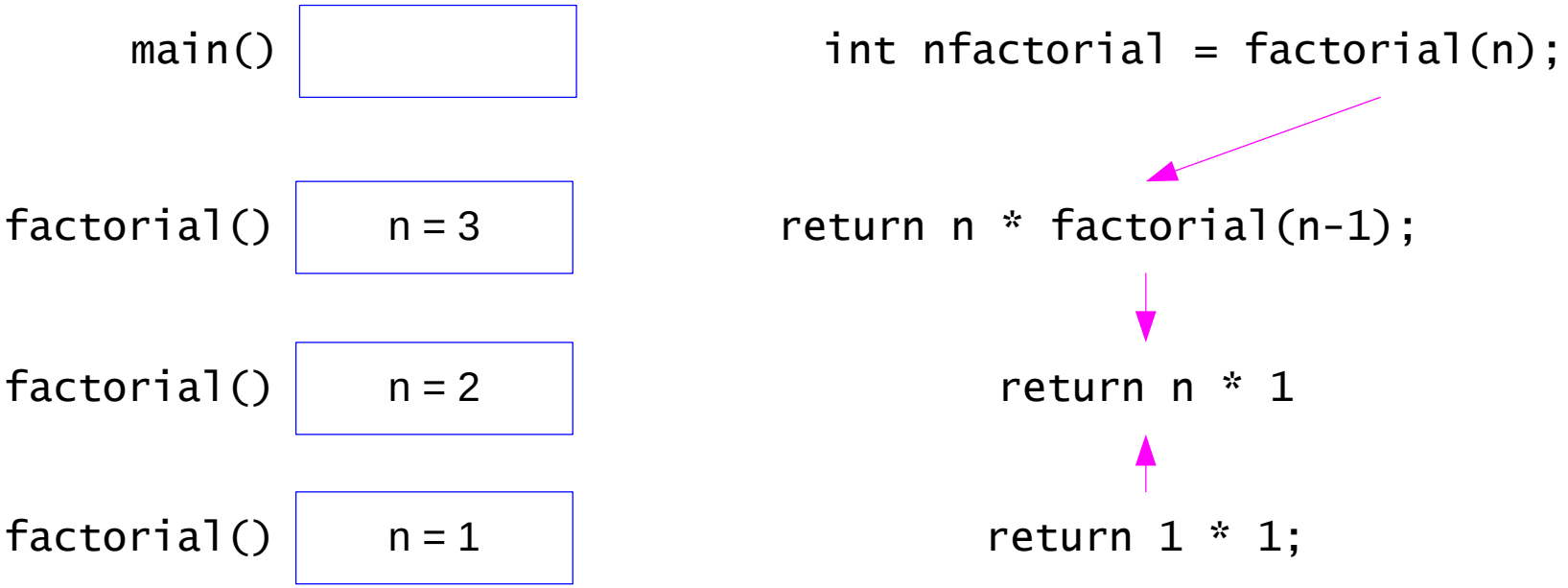

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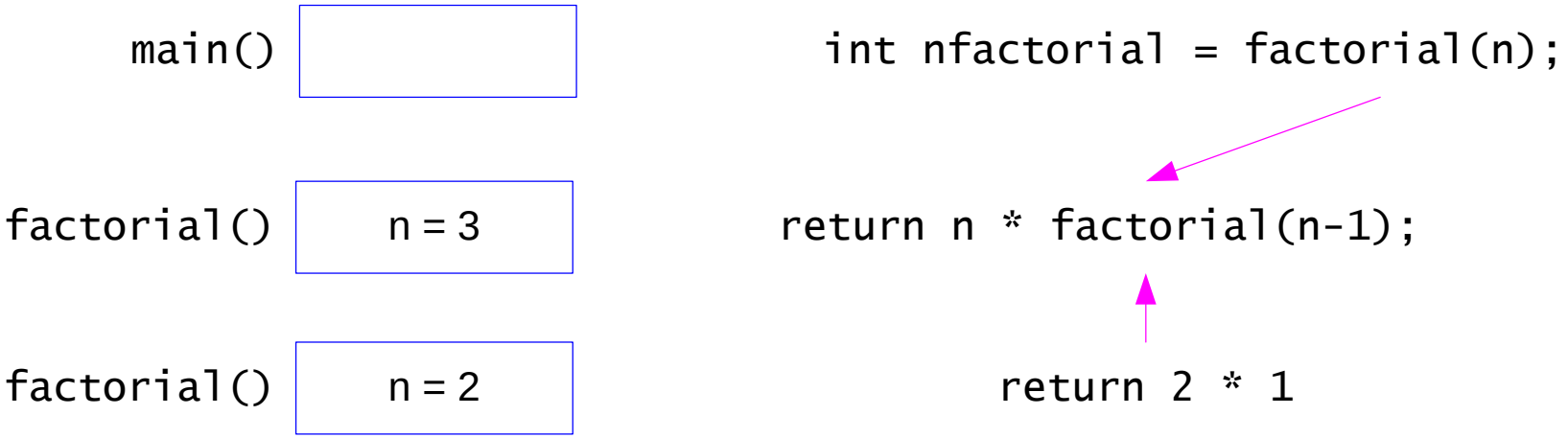

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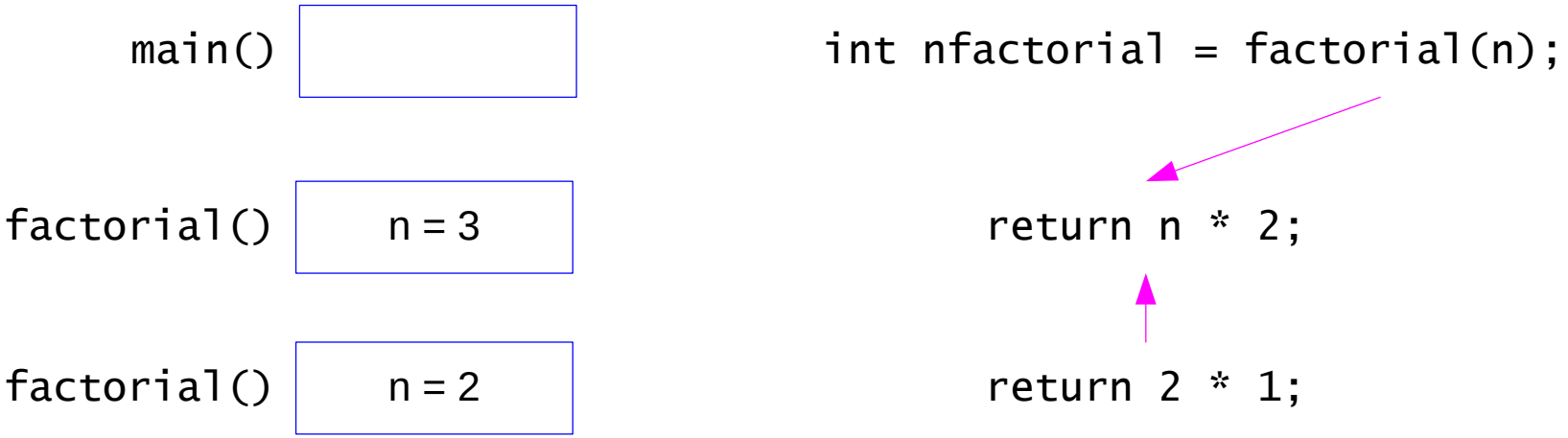

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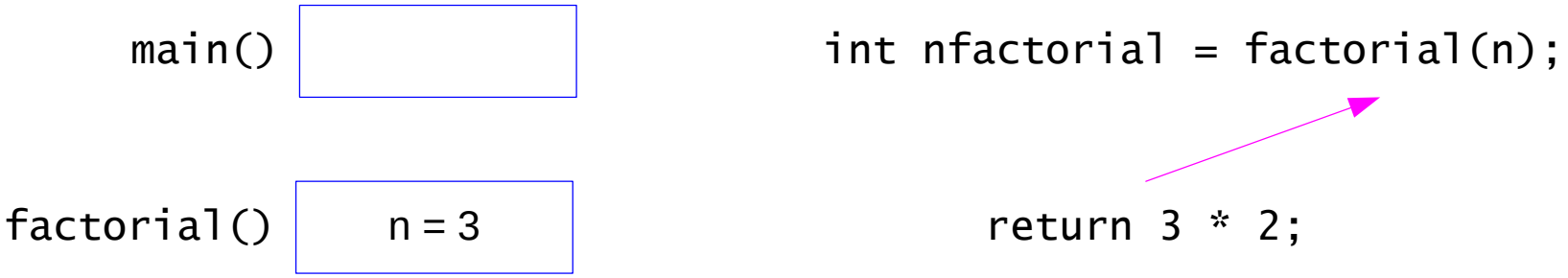

Recursive invocation

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Infinite recursion

- A common programming error when using recursion is to not stop making recursive calls.
  - The program will continue to recurse until it runs out of memory.
  - Be sure that your recursive calls are made with simpler or smaller subproblems, and that your algorithm has a base case that terminates the recursion.

Fibonacci Series Code

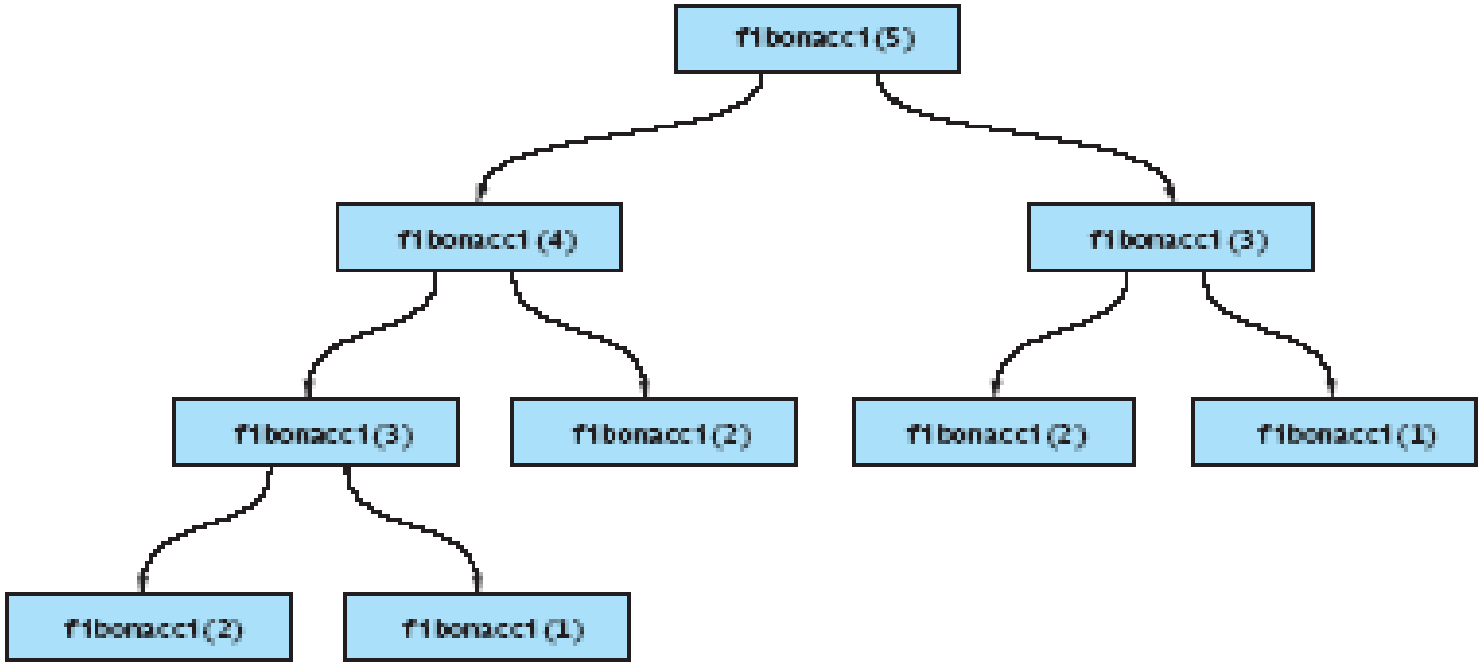
```
public static int fib (int n) {
  if (n <= 2)
    return 1;
  else
    return fib(n-1) + fib(n-2);
}
```

This is straightforward, but an inefficient recursion ...

1, 1, 2, 3, 5, 8, 13,.....

Efficiency of Recursion: Inefficient Fibonacci

FIGURE 7.6
Method Calls Resulting from `fibonacci(5)`



Efficient Fibonacci

- ***Strategy:*** keep track of:
  - *Current* Fibonacci number
  - *Previous* Fibonacci number

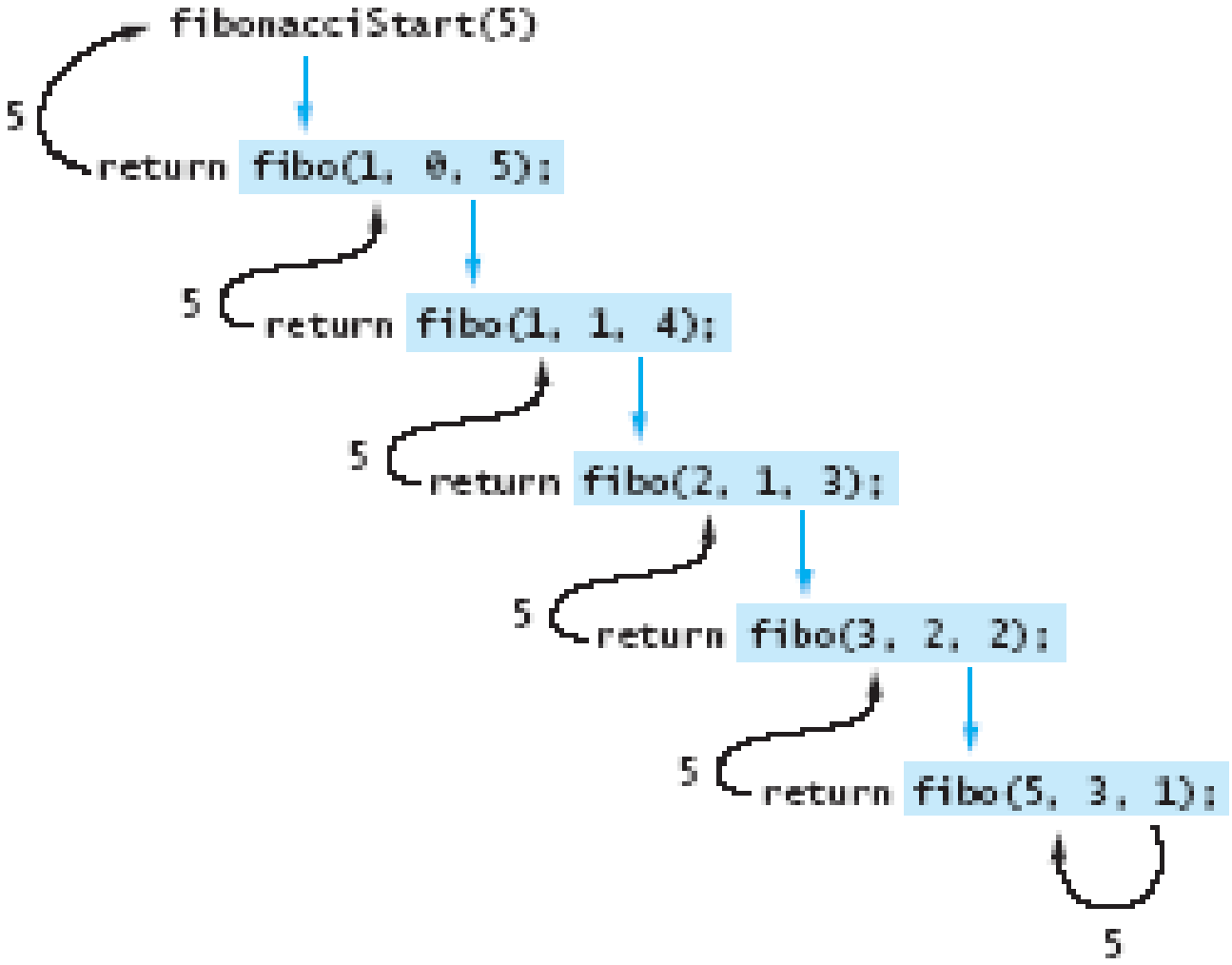
Efficient Fibonacci: Code

```
public static int fibStart (int n) {
   return fibo(1, 1, n);
}

private static int fibo (
     int curr, int prev, int n) {
   if (n <= 1)
     return curr;
   else
     return fibo(curr+prev, curr, n-1);
}
```

Efficient Fibonacci: A Trace

FIGURE 7.7
Trace of `fibonacciStart(5)`



Recursion Advantages

- Expressive Power
  - Recursive code is typically much shorter than iterative.
- More appropriate for certain problems.
- Intuitive programming from mathematic definitions.

Recursion Disadvantages

- Usually slower due to call stack overhead.
- Faulty Programs are very difficult to debug.
- Difficult to prove a recursive algorithm is correct

End